Md Shibly Sadique

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Education

2018-2027	Ph.D., Electrical & Computer Engineering, Old Dominion University, USA *Expected graduation: July 2027* *Dissertation: Computational Modeling for Classification and Survivability Prediction in Recurrent Brain Tumor, Integrating Biophysical Tumor Growth Models (PDE)* *Advisor: Dr. Khan M. Iftekharuddin*
2014-2016	M.Sc., Electrical & Electronic Engineering, University of Dhaka, Bangladesh *Thesis: Design, Simulation, and Measurement of a Slot-Loaded Microstrip Patch Antenna* *Advisor: Dr. Anis Ahmed*
2010-2014	B.Sc., Electrical & Electronic Engineering, University of Dhaka, Bangladesh

Research Areas

Biomedical imaging	Multimodal and 3D volumetric MRI, functional MRI, and CT; registration, segmentation, and classification; physics-informed (PDE) modeling, radiomic feature extraction, longitudinal disease progression modeling, uncertainty quantification, and explainable AI.
Computer vision	Deep learning and classical image processing for biological structure detection; automated segmentation and predictive modeling with U-Net, nnU-Net, neural ODEs, uncertainty-aware frameworks, and generative models.
Data mining	Survival and risk prediction from imaging biomarkers; prognostic modeling; inferential statistical modeling for interpretability; feature selection; class-imbalanced data analysis.

Peer-Reviewed Publications

Journal Articles

- [J1] Temtam, A., Witherow, M.A., Ma, L., **Sadique, M.S.**, Moeller, F.G., & Iftekharuddin, K.M. (2025). *Functional Brain Network Identification in Opioid Use Disorder Using Machine Learning Analysis of Resting-State fMRI BOLD Signals. Computers in Biology and Medicine*, 196(C), 110946.
- [J2] Temtam, A., Witherow, M.A., Ma, L., **Sadique, M.S.**, Moeller, F.G., Kenneth, C., Wright, D., & Iftekharuddin, K.M. (2025). *Differentiating Opioid Use Disorder from Healthy Controls via ML Analysis of rs-fMRI Networks. Journal of Clinical and Translational Science*, 9(s1), 110-111.
- [J3] **Sadique, M.S.**, Farzana, W., Temtam, A., Lappinen, E., Vossough, A., & Iftekharuddin, K.M. (2024). *Brain Tumor Recurrence vs. Radiation Necrosis Classification and Patient Survivability Prediction. IEEE Journal of Biomedical and Health Informatics.*
- [J4] Ohi, M.A.R., **Sadique, M.S.**, Hussain, S., & Ahmed, A. (2017). *Design and Fabrication of Slot-Loaded Microstrip Patch Antenna at 2.45 GHz. International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering.*

Conference Proceedings

- [C1] Farzana, W., Witherow, M.A., Longoria, I., **Sadique, M.S.**, Temtam, A., & Iftekharuddin, K.M. (2024).

Domain Adaptive Federated Learning for Multi-Institution Molecular Mutation Prediction and Bias Identification. In Proc. SPIE Medical Imaging: Computer-Aided Diagnosis, vol. 12927, 368-375.

[C2] **Sadique, M.S.**, Farzana, W., Temtam, A., & Iftekharuddin, K.M. (2023). *Class Activation Mapping and Uncertainty Estimation in Multi-Organ Segmentation.* In Proc. SPIE Medical Imaging: Computer-Aided Diagnosis, vol. 12465, 169-174.

[C3] Temtam, A., Ma, L., Moeller, F.G., **Sadique, M.S.**, & Iftekharuddin, K.M. (2023). *Opioid Use Disorder Prediction Using Machine Learning of fMRI Data.* In Proc. SPIE Medical Imaging: Computer-Aided Diagnosis, vol. 12465, 142-146.

[C4] **Sadique, M.S.**, Temtam, A., Lappinen, E., & Iftekharuddin, K.M. (2022). *Radiomic Texture Feature Descriptor to Distinguish Recurrent Brain Tumor from Radiation Necrosis Using Multimodal MRI.* In Proc. SPIE Medical Imaging: Computer-Aided Diagnosis, vol. 12033, 668-674.

Book Chapters and Workshop Papers

[B1] **Sadique, M.S.**, Rahman, M.M., Farzana, W., Glandon, A., Temtam, A., & Iftekharuddin, K.M. (2023). *Brain Tumor Segmentation: Glioma Segmentation in Sub-Saharan Africa Patients Using nnU-Net.* In Int'l Challenge on Cross-Modality Domain Adaptation for Medical Image Segmentation, 322-331. Springer.

[B2] Temtam, A., **Sadique, M.S.**, Rahman, M.M., Farzana, W., & Iftekharuddin, K.M. (2023). *Pediatric Brain Tumor Segmentation Using Multiresolution Fractal Deep Neural Network.* In Int'l Challenge on Cross-Modality Domain Adaptation for Medical Image Segmentation, 332-340. Springer.

[B3] **Sadique, M.S.**, Rahman, M.M., Farzana, W., Glandon, A., Temtam, A., & Iftekharuddin, K.M. (2023). *Local Synthesis of Healthy Brain Tissue Using an Enhanced 3D Pix2Pix Model for Medical Image Inpainting.* In Int'l Challenge on Cross-Modality Domain Adaptation for Medical Image Segmentation, 312-321. Springer.

[B4] **Sadique, M.S.**, Rahman, M.M., Farzana, W., Temtam, A., & Iftekharuddin, K.M. (2022). *Brain Tumor Segmentation Using Neural Ordinary Differential Equations with UNet-Context Encoding Network.* In Int'l MICCAI Brainlesion Workshop, 205-215. Springer.

[B5] Rahman, M.M., **Sadique, M.S.**, Temtam, A.G., Farzana, W., Vidyaratne, L., & Iftekharuddin, K.M. (2021). *Brain Tumor Segmentation Using UNet-Context Encoding Network.* In Int'l MICCAI Brainlesion Workshop, 463-472. Springer.

[B6] Farzana, W., Temtam, A.G., Shboul, Z.A., Rahman, M.M., **Sadique, M.S.**, & Iftekharuddin, K.M. (2021). *Radiogenomic Prediction of MGMT Using Deep Learning with Bayesian Optimized Hyperparameters.* In Int'l MICCAI Brainlesion Workshop, 357-366. Springer.

Preprint

[P1] Kofler, F., Meissen, F., Steinbauer, F., Graf, R., Oswald, E., de la Rosa, E., Li, H.B., Baid, U., Hoelzl, F., Turgut, O., **Sadique, M.S.**, et al. (2023). *The Brain Tumor Segmentation (BraTS) Challenge 2023: Local Synthesis of Healthy Brain Tissue via Inpainting.* arXiv: 2305.08992.

Technical Presentations

2026 — *Adult-to-Pediatric Domain Shift in Brain Tumor Segmentation Using Multiresolution Fractal Deep Neural Networks.* SPIE Optics + Photonics, San Diego, California.

2023 — *Class Activation Mapping and Uncertainty Estimation in Multi-Organ Segmentation.* SPIE Medical Imaging: Computer-Aided Diagnosis, San Diego, California.

2022 *Brain Tumor Segmentation Using Neural Ordinary Differential Equations with UNet-Context Encoding Network.* Int'l MICCAI Brainlesion Workshop.

2022 *Radiomic Texture Feature Descriptor to Distinguish Recurrent Brain Tumor from Radiation Necrosis Using Multimodal MRI.* SPIE Medical Imaging: Computer-Aided Diagnosis.

Research Experience

2018-Present **Graduate Research Assistant**, Vision Lab
Old Dominion University, Norfolk, Virginia; advised by Dr. Khan M. Iftekharuddin

2025-Present *Quantifying Learnability in Brain Tumor Growth Modeling Under Sampling Noise*

- Examined how sampling noise constrains learnability in PDE-based tumor growth models.
- Proposed the Resolvable Expressive Capacity framework with eigentask decomposition to quantify reliably learnable growth directions under uncertainty.
- Developing a Bayesian inference framework for noise-aware digital twin models of tumor evolution.

2023-2026 *Adult-to-Pediatric Domain Shift in Brain Tumor Segmentation*

- Developed a multiresolution fractal deep neural network with gradient-reversal domain alignment, transferring from a large adult glioma cohort to a substantially smaller pediatric target cohort.
- Applied Monte-Carlo dropout for voxel-level epistemic uncertainty, flagging low-confidence regions for radiologist review.

2021-2025 *Functional Brain Network Identification in Opioid Use Disorder* Project: NIH

- Extracted time-frequency features from resting-state fMRI BOLD signals to localize aberrant neural activity within functional networks.
- Applied Random Forest modeling with Boruta selection and statistical analysis to identify the functional networks with the highest discriminative power for opioid use disorder against healthy controls.

2021-2024 *Noninvasive Classification and Prognostic Modeling of High-Grade Brain Tumors: Recurrence and Radiation-Induced Necrosis* Project: NIH/NIBIB

- Developed a computational model addressing the clinical difficulty of separating recurrent brain tumor from radiation necrosis, which present near-identical imaging features, using multiresolution radiomic features with statistically rigorous repeated random sub-sampling.
- Achieved an area under the curve of 0.892 ± 0.055 in classifying recurrence against necrosis.
- Designed a copula-based survival pipeline addressing dependent censoring in glioblastoma, reaching cross-validated AUC ≈ 0.77 and outperforming Cox regression with clearer stratification of patient risk groups.

2021-2023 *Uncertainty-Aware Multi-Organ Segmentation Using MRI and CT*

- Proposed an uncertainty quantification framework using ensemble nnU-Nets with 5-fold cross-validation, enabling detection of low-confidence predictions.
- Quantified predictive reliability by combining total uncertainty (entropy) with epistemic uncertainty (mutual information) across ensemble models.
- Demonstrated that uncertainty estimates reliably stratify erroneous segmentations, improve interpretability, and enable targeted clinical error analysis.

2018-2023 *Brain Tumor Segmentation and Validation in MRI* Project: NIH/NIBIB

- Developed UNCE-NODE, a neural ODE-enhanced U-Net for multimodal MRI brain tumor segmentation, introducing continuous-depth modeling with shared weights across integration steps.
- Reduced trainable parameters by roughly 30% and accelerated inference while preserving Dice accuracy.
- Applied uncertainty quantification to assess prediction confidence, improving the reliability and interpretability of automated segmentation.
- Participated three times in the global challenge organized by the MICCAI society.

2014-2016 **Research Assistant**, Microwave & Optical Fiber Communications Lab
University of Dhaka, Bangladesh; advised by Dr. Anis Ahmed

- Built simulation capability, fabrication, and experimental setup for microstrip patch antenna research in the Department of Electrical & Electronic Engineering.

Teaching Experience

2021-Present **Teaching Assistant**, Old Dominion University
ECE 481W Preparatory ECE Senior Design; ECE 487 ECE Senior Design II

- Worked with the course instructor to develop course materials and synchronize related class modules through Canvas.
- Prepared ABET outcome reports.
- Graded quizzes, assignments, and projects, and returned grades and feedback through Canvas.

2019-Present **Teaching Assistant**, Old Dominion University
ECE 201 Circuit Analysis I and ECE 202 Circuit Analysis II (Fall 2019, Spring 2020); ECE 302 Linear System Analysis (Fall 2020); ECE 287 Fundamental Electric Circuit Laboratory (Summer 2025)

- Conducted problem-solving classes on assignments and exam preparation, and led laboratories and tutorials.
- Designed MATLAB/Simulink and Multisim exercises linking circuit theory to applications.
- Proctored examinations and graded quizzes, assignments, projects, and exam scripts.

2017 **Course Instructor**, Dhaka International University, Bangladesh
ECE 201 Circuit Analysis I and Fundamental Electric Circuit Laboratory (Spring 2017, Fall 2017)

- Taught theory classes with laboratory experimentation; set class tests and final examinations and evaluated all scripts.
- Prepared and delivered pre-laboratory lectures, set up and demonstrated experiments, and graded weekly performance.

Mentorship

Summer 2023 Undergraduate researcher, NSF-REU Program, Old Dominion University
Privacy-Preserving Federated Learning with Domain Alignment for Multi-Site Molecular Mutation Prediction from Radiomic Features

Summer 2022	Undergraduate researcher, NSF-REU Program, Old Dominion University *Using a Privacy-Preserving Medical Image Analysis Framework to Predict Prognosis of Patients with Gliomas*

Awards and Honors

2023	SPIE Medical Imaging Student Conference Support
2022	SPIE Medical Imaging Student Conference Support
2015	National Science, Information and Communication Technology (NSICT) Fellowship for M.S. Thesis, Bangladesh

Research Competitions

2022	Team Lead, Multi-modal Abdominal Multi-Organ Segmentation Challenge (AMOS) *Eighth place in Task 1 and Task 2*
2021-2026	Participant, RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge

Academic and Professional Service

Journal	Reviewer, *Franklin Open* (Elsevier)
Conference	Reviewer, Joint Int'l Conference on Informatics, Electronics & Vision (ICIEV); Int'l Conference on Imaging, Vision & Pattern Recognition (ICIVPR)

Professional Workshops and Training

May 2025	Co-clinical Imaging Research Resource Program (CIRP) Annual Virtual Meeting
May 2023	Medical Imaging De-Identification, National Cancer Institute (NCI)
May 2022	DeapSECURE Summer Workshop Series: big data, machine learning, and deep learning with neural networks. NSF-funded training for REU students; served as volunteer tutor.

Technical Skills

Programming	Python, MATLAB, R (proficient); C, shell script (basic)
Deep learning	CNN, RNN, LSTM, U-Net, nnU-Net, Neural ODEs, generative AI, diffusion models, transformers, attention
Machine learning	Classification, statistical analysis, predictive modeling, feature selection and engineering, correlation analysis, optimization, transfer learning, domain adaptation, supervised and unsupervised learning, dimensionality reduction, clustering, class balancing, naïve Bayes, gradient boosting, random forests, SVM, PCA, LDA
Imaging software	MIPAV, FSL, ImageJ, FreeSurfer, 3D Slicer, ANTs, SimpleITK, NiBabel, ITK-SNAP
Libraries & frameworks	TensorFlow, PyTorch, Keras, scikit-learn, pandas, NumPy, Matplotlib, OpenCV, Docker, MedPerf, CUDA, GPU, Slurm (HPC)

Graduate Coursework

Engineering	ECE 601 Linear Systems; ECE 607 Machine Learning 1; ECE 611 Numerical Methods in Engineering Analysis; ECE 612 Digital Signal Processing 1; ECE 883 Digital Image Processing; ECE 887 Digital Communications
Computing	CS 822 Machine Learning; CS 823 Introduction to Bioinformatics; CS 895 3D Image Deep Learning; CS 895 Practical Machine Learning Application; STAT 697 Design and Experimental Analysis

Professional Memberships

Current	IEEE Graduate Student Member; SPIE Student Member

References

Available upon request.